Afaf Binta Shamsuddin

Mirpur, Dhaka +8801952209233 afaf.binta.shamsuddin@gmail.com
afafbintashamsuddin AfafBintaShamsuddin

Education

Bachelor of Science in Computer Science 2022 – Present
BRAC University *Dhaka, Bangladesh*
- Relevant Coursework: DBMS, System Design, Networking, French, German

Higher School Certificate (HSC), Science 2018 – 2020
Shaheed Bir Uttam Lt. Anwar Girls' College *Dhaka, Bangladesh*
- GPA: **5.00**/5.00

Secondary School Certificate (SSC), Science 2016 – 2018
Shaheed Bir Uttam Lt. Anwar Girls' College *Dhaka, Bangladesh*
- GPA: **5.00**/5.00
- Achived Best Student Performance Award 2018 based on merit.

Experience

English Tutor (Cambridge Curriculum) 2023 – Present
Desh Tutor *Dhaka, Bangladesh*
- Tutored **3 students (KG–Class 6)** under the Cambridge curriculum, **6+ hours weekly**, guiding one student from Class 2 to Class 5 with up to **40% improvement in Mathematics** scores.
- Designed custom question sets and study guidelines tailored to each student's level, and administered periodic assessments to track progress.

Extracurricular Activities

Senior Executive 2022 – 2025
BRAC University Art & Photography Society *Dhaka, Bangladesh*
- Managed **5+ events and exhibitions**, including the photography exhibition **"Carpe Diem."**
- Captured and edited photography for **9+ events**.

General Secretary 2016 – 2017
SAGC Science Club *Dhaka, Bangladesh*
- Organized **2+ national science festivals** with **1,000+ participants**, overseeing end-to-end event execution.
- Led and coordinated a team of **20+ volunteers**, directing t-shirt distribution and managing participant registration.

Skill

OS: Linux, Windows
Languages & Tools: Python, MySQL, Wireshark, Packet Tracer
Software: Microsoft Office Suite (Word, Excel, PowerPoint, Outlook), GitHub
Soft Skills: Leadership, Team Management, Communication, Problem Solving, Time Management, Adaptability, Critical Thinking, Collaboration

Language Proficiency

Bangla: Native Speaker | **English:** IELTS Band Score **6.5**
German: Completed Language Course (Certified) | **French:** Completed Language Course (Certified)

Projects

AdAstra (Astronomical Image Classification & RAG Platform) *(Team of 4)* 2026
PyTorch, EfficientNet-B0, CLIP, FAISS, Gemma, Vercel, MongoDB Atlas Live
- Built a two-stage CNN recognition pipeline (EfficientNet-B0) classifying astronomical images into **5 object types** and galaxy morphology, reaching **83.7% test accuracy** on SpaceNet and **86.9%** on Galaxy Zoo across **70,000+ training images**, with a CLIP-based gate rejecting non-astronomical inputs and a confidence rule flagging uncertain predictions.
- Designed a retrieval-augmented explanation engine: a FAISS index over **6,222 chunks** from 113 curated Wikipedia articles feeds the Gemma language model to generate citation-grounded, multi-level (beginner/intermediate/advanced) explanations, with post-generation citation verification to strip unsupported references.
- Shipped a full-stack web platform (Vercel backend API, MongoDB Atlas) with authenticated accounts, saved analysis history, a source-cited astronomy Q&A assistant, and a sky explorer / Moon-phase page, serving concurrent requests via a load-balanced architecture.

Achievements

DRMC-Savlon National Science Carnival X (2017) — **1st Place**, Project Display Competition, secondary level, organized by Dhaka Residential Model College.
MACES Photography Contest (2024) — **Winner**, selected among **12 winners** from **1000+** of participants.